WEEK 4 · LESSON 20 OF 24 · HANDOUT

Explainable AI

Study time: **about 40 minutes** (reading, video and activity)

▶ **Watch:** https://www.youtube.com/watch?v=yJkCuEu3K68 — open the lesson online to watch inside the course.

Your goal in this lesson: Explain why some AI models are "black boxes", why explanations matter, and describe common ways to explain a model's decisions.

The black-box problem

A decision tree can show its reasoning: *sales above 100,000 RWF → few late payments → approve*. A deep neural network with millions of weights cannot. It may be very accurate, but nobody can easily say **why** it made a particular decision. Such models are called **black boxes**.

This matters when a decision affects someone's life. If a bank's AI refuses your loan, you deserve to know why — and the bank needs to be sure the reason is legitimate, not a hidden bias.

Why explanations matter

Who	Why they need explanations
The affected person	To understand, challenge or fix the decision ("increase my savings history")
The organisation	To trust the model, catch mistakes and meet legal duties
Engineers	To debug the model and find data problems or leakage
Regulators	To check the system is safe and fair

Interpretable models vs. explaining black boxes

There are two approaches:

1. **Use an interpretable model** — linear or logistic regression, small decision trees. You can read the logic directly. Often good enough, especially for tabular data.
2. **Explain a complex model after training** — use tools that describe what the black box is doing.

A good rule: **start with the simplest model that works.** Use a black box only when it is clearly better, and then add explanation tools.

Common explanation methods

- **Feature importance** — which features matter most to the model overall. "Repayment history is the strongest factor, then monthly sales."
- **Local explanations** — why *this one* decision was made. Tools such as **LIME** and **SHAP** estimate how much each feature pushed a single prediction up or down.
- **Counterfactual explanations** — what would need to change for a different result. "If your monthly sales were 30,000 RWF higher, the loan would be approved." These are often the most useful for ordinary people.

- **Saliency maps** — for images, highlight which pixels the model focused on. A crop-disease model should focus on the leaf spots, not the background soil.

Explanations can reveal hidden problems

A famous lesson from research: a model that seemed to tell wolves from huskies was really looking at **snow in the background** — most wolf photos had snow. Only an explanation method revealed the shortcut. Explanations help catch models that are right for the wrong reasons.

Limits of explanations

Explanation tools give **approximations**, not the full truth of a complex model. Different tools can disagree. Explanations support human judgement; they do not replace it.

Remember this: Black-box models can be accurate but hard to understand. Prefer interpretable models where possible; otherwise use feature importance, local explanations and counterfactuals — and remember they are approximations.

Name: _____ Date: _____

Activity

Task 1

Imagine an AI rejects a student's scholarship application. Write the explanation the student should receive, in plain words, including a counterfactual ("If… then…").

Task 2

A crop-disease model's saliency map shows it focuses on the soil, not the leaf. What could be wrong with the training data? Suggest one fix.

Self-check

Think about your answers here, then take the quiz online — it is marked instantly and explains every answer.

1. What is a 'black-box' model?
   - A. A model whose reasons for a decision are hard to see or explain
   - B. A model stored in a black case
   - C. A model with no data
   - D. A model that only works at night
2. Which model is usually easiest to interpret directly?
   - A. A random mix of models
   - B. A small decision tree
   - C. A large language model
   - D. A deep neural network with millions of weights
3. 'If your monthly sales were 30,000 RWF higher, the loan would be approved.' This is a…
   - A. Counterfactual explanation
   - B. Loss function
   - C. Saliency map
   - D. Confusion matrix
4. What do LIME and SHAP help with?
   - A. Explaining how features pushed one particular prediction
   - B. Storing passwords
   - C. Collecting data
   - D. Training faster

5. A wolf-vs-husky model was really detecting snow. What does this show?

A. Explanations can reveal a model that is right for the wrong reasons

B. Wolves only live in snow

C. Snow is a good feature

D. Models cannot be explained

6. What is an important limit of explanation tools?

A. They give approximations and can disagree

B. They are always 100% exact

C. They only work on text

D. They replace all human judgement

© WAMUGII TECH SOLUTIONS · wamugii.kento.rw · Original course material — for enrolled students.